BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY

Bachelor's Thesis Assignment

172352

Institut:	Department of Computer Graphics and Multimedia (DCGM)
Student:	**Helešic Adam**
Programme:	Information Technology
Title:	**Advanced Injury Simulation of Game Characters**
Category:	Computer Graphics
Academic year:	2025/26

Assignment:

1. Familiarize yourself with game development in the Godot engine.
2. Study the issue of game character animation and damage during combat.
3. Research existing and possible solutions to the given problem.
4. Design a user interface and the final functionality of an application that demonstrates advanced character reactions to injuries.
5. Implement the application and demonstrate its use with different types of data.
6. Perform user or performance testing of the results.
7. Document and publish the final application.
8. Create a video representing the results of your work.

Literature:

- Adams, Ernest, and Joris Dormans. Game mechanics: advanced game design. New Riders, 2012. ISBN 0321820274, 9780321820273.
- Geijtenbeek, T., Vasilescu, D., Egges, A. Injury Assessment for Physics-Based Characters. In: Allbeck, J.M., Faloutsos, P. (eds) Motion in Games. MIG 2011. Lecture Notes in Computer Science, vol 7060. Springer, Berlin, Heidelberg. 2011. DOI: 10.1007/978-3-642-25090-3_7.
- Francis Laclé, Nicolas Pronost. Real-Time Musculoskeletal Model for Injury Simulation on 3D Human Characters. Motion In Games Conference (MIG), Nov 2012, Rennes, France.
- Aristidou, A., Lasenby, J., Chrysanthou, Y. and Shamir, A. Inverse Kinematics Techniques in Computer Graphics: A Survey. Computer Graphics Forum, 37: 35-58. 2018. DOI: 10.1111/cgf.13310.

Requirements for the semestral defence:
Items 1-4, experiments leading to item 5.

Detailed formal requirements can be found at https://www.fit.vut.cz/study/theses/

Supervisor:	**Chlubna Tomáš, Ing., Ph.D.**
Head of Department:	Černocký Jan, prof. Dr. Ing.
Beginning of work:	1.11.2025
Submission deadline:	13.5.2026
Approval date:	7.11.2025